

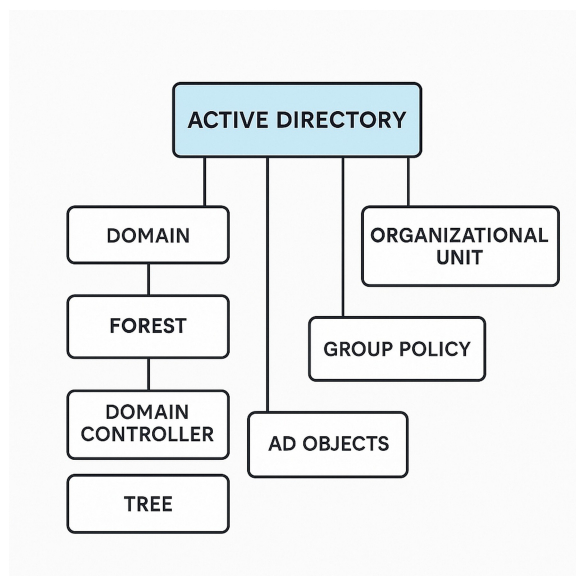
What is Active Directory?

Active Directory (AD) is a **Microsoft service** that helps manage users, computers, and resources in a **Windows-based network**.

Think of it like a **digital office directory** — managing employees (users), rooms (computers), access cards (permissions), and departments (groups/OU).

Main Components of Active Directory

Component	Description
Domain	A logical group of users, computers, and resources managed as a unit
Domain Controller (DC)	A server that stores a copy of the AD database and authenticates users
Forest	A collection of one or more domains that share a common schema
Tree	A group of domains in a hierarchical structure (child-parent relationship)
Organizational Unit (OU)	A container used to organize users, computers, or groups for easy management
Group Policy (GPO)	Set of rules used to control user and computer settings across the network
AD Objects	Anything stored in AD (users, computers, groups, printers, etc.)
Sites	Represents the physical structure (e.g., offices in different cities)
Schema	Blueprint of what AD can store and how (attributes, classes, rules)
Global Catalog	A searchable index of all AD objects (helps with logins and searches)



What is an Organizational Unit (OU)?

- **OU** is a container within a domain.
- Used to **group users, computers, or groups** for easier management.

- **You can apply different Group Policies** to different OUs.

🧠 Example:

In a college network:

Domain: codingseekho.com

└— OU: Students

└— OU: Teachers

└— OU: Admin Staff

Each OU can have different rules and permissions.

🌱 What is Group Policy?

🎯 Purpose:

Group Policy allows admins to define **rules/settings for users and computers** centrally.

Can Control:

Disable Control Panel

Set desktop wallpaper

Lock screen after 5 min

Disable USB ports

Software install/remove

📌 GPO = Group Policy Object

- A set of settings stored in AD
- Can be applied at **Site, Domain, or OU level**

🔄 How It All Works Together (Example):

1. User logs in from a computer inside the domain.
2. Domain Controller checks username & password.
3. If valid, user is allowed access.
4. Group Policy settings are applied (e.g., no USB, set wallpaper).
5. User accesses network printers, files, and apps based on permission.

📄 Summary Table

Term	Meaning
Active Directory	Microsoft service for managing users, computers, groups
OU (Organizational Unit)	Logical container for organizing AD objects
Group Policy	Rules and settings applied to users or computers
Domain	A group of objects under the same AD structure
Forest	One or more domains sharing schema and configuration
Domain Controller	Server that stores and manages the AD database

Real-World Analogy

AD Element	Real-Life Example
Domain	Your office/company
OU	Departments like HR, IT, Sales
User Object	Employees
Group Policy	Office rules (e.g., no phone in office)
Domain Controller	Security guard who checks ID and allows entry